

Element J: Documentation of External Evaluation

Student E, a student in our engineering class, needs a way to get rid of wasps, hornets, and other insects that enter his room at night when he sleeps. The student's bedroom has a serious insect problem. The insects make the space noisy and affect the students' ability to rest.

The expert we chose to evaluate our testing procedures was Ms. Z. She reviewed our testing plan and made important improvements. She adjusted the order of the testing plan, which will prevent confusion when conducting the tests. She also encouraged adding important criteria, such as the maximum volume of the prototype, and testing in many locations for the most accurate results.

The experts we chose to evaluate our current prototype were Dr. H. and Dr. M. We chose Dr. H. due to her background in biology and co-authoring of previous biology papers on insects. Dr. M. was chosen as he is an avid beekeeper and has experience with preventing insects from entering houses. We received critical feedback from both Dr. H and Dr. M. Dr. H. informed us that the flashlight that was used as a lure does not attract the insects, but instead blinds them and leaves them confused, resulting in the illusion of them being "attracted" to the light. She recommended we use a fragrance or pheromone to attract the insects. Dr. M. had an interesting insight into how the insects may be entering the house. He said that houses with eaves, overhanging roofs, often have a continuous vent running along the house. The vent may have a small hole in it, the insects might enter Student E's room from that hole. Student E provided feedback that the insect-capturing device is more successful at capturing insects outside of his house than inside his room.

Our feedback from our experts, Dr. H, Dr. M, Ms. Z, and Student E, was crucial in the development of prototype 2. Instead of solely relying on the flashlight as a lure, we decided to

add a projector lens to disperse the light and started using a pheromone or scent attractant. The light will serve as part of the trapping mechanism rather than a lure.